

MARY CLAYTON SOTO

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EDUCATION

Duke University, *Pratt School of Engineering*

Aug 2024 – B.S. 2028

- **Majors:** Biomedical Engineering & Mathematics; **Minor:** Visual Arts
- **GPA:** 3.92/4.00
- **Activities and Societies:** Club Soccer, Theta Tau Engineering Fraternity, Phi Alpha Delta, Tri Delta
- **Relevant Coursework:** Quantitative Physiology with Biostatistical Applications, Engineering Design and Collaboration, Mechanics of Solids, Python Programming for Math, Signals and Systems, Modeling Cellular and Molecular Systems

Maclay School, *Salutatorian*

May 2024

ENGINEERING EXPERIENCE

Bass Connections Research Program, *Developing Surgical Education Models Team Member*

Mar 2025 – Present

- Collaborate with Duke's SEAL Lab to design and refine surgical education models that improve efficiency, accessibility, and consistency in clinical training.
- Design and iterate hydrogel organ models using MRI-derived DICOM scans and 3D Slicer segmentation, creating Fusion 360 molds and tuning materials to replicate human tissue mechanics for surgical simulation.
- Contribute to development and validation of a computer-vision knot-tying assessment system, using video processing to quantify performance and support objective surgical skill evaluation.

FSU Microparticle Lab, *Research Assistant*

May – Aug 2025

- Developed a MATLAB pipeline to segment and classify fluorescence cell images, analyzing ~20,000 cells and generating a standardized dataset to support a manuscript published in ACS Applied Materials & Interfaces.
- Improved classification accuracy and consistency by optimizing convolution-based segmentation and thresholding across heterogeneous fluorescence channels.

Tallahassee Research Institute, *Research Coordinator*

May 2026 – Present

- Manage clinical research workflows and patient data for cardiovascular studies involving novel medical devices and research protocols.
- Compile and analyze patient data into structured reports, translating complex findings into actionable insights to support data-driven biomedical research.

Victor Tech, *Intern*

Jan – May 2024

- Designed and fabricated custom prosthetics using 3D printing, iterating on form and functionality to meet individual clinical needs.

ACADEMIC PROJECTS

Rollator Design Project

Aug 2024 – May 2025

- Designed and prototyped a device to secure an oxygen canister to Duke Hospital rollators, improving patient mobility, safety, and clinical workflow efficiency.
- Conducted user-centered design analysis within hospital constraints and evaluated safety factors and customer criteria to ensure reliable performance in a clinical environment.

Autonomous Arduino Robot

Jan – May 2025

- Built and programmed an Arduino-based robot using TI path sensors and IR thermal sensing, integrating multiple hardware components and developing control logic for autonomous navigation and wireless communication.

Yeast Genomic Library Project

Aug – Dec 2025

- Executed a recombinant DNA workflow including yeast genomic DNA isolation, plasmid cloning, bacterial transformation, and validation via PCR, gel electrophoresis, Sanger sequencing, and bioinformatic analysis.

Data Analysis Project

Aug – Dec 2024

- Developed a Python-based analysis workflow to extract pixel data from environmental maps, compute quadrant-level metrics, and perform regression modeling to assess relationships between green space, land use, and urban temperature.

LEADERSHIP & INVOLVEMENT

EGR105: Coding for Engineers, *Teaching Assistant*

Aug 2025 – Present

- Mentor first-year engineering students in Python algorithm design, reinforcing debugging workflows and Git-based version control.
- Teach use of NumPy and Matplotlib for data analysis and visualization, emphasizing efficient data handling and interpretation.

Theta Tau, *Vice President of Professional Development*

Aug 2025 – Present

- Execute events to highlight engineering research and innovation, expanding members' professional opportunities and skills.
- Created a centralized member resource database for academic planning, networking, career development.

Lifeguard, *City of Tallahassee*

May – Aug 2023, 2024, 2025

- Supervised high-traffic pool operations, maintaining rigorous safety standards and emergency readiness in time-sensitive situations.

SKILLS

Python & MATLAB (*Data Analysis, Image Processing & Segmentation, Algorithm Design*), Prototyping & Fabrication (*3D Printing, Laser Cutting, Mold-Making, Casting*), Statistical Analysis (*Correlation Studies, Data Synthesis, ANOVA Tests*), Leadership & Mentorship (*Teaching Assistant, Tutoring*), Communication & Organization (*Event Planning, Outreach, Public Speaking*)